

Cylvero system map

Everything the cylinder app is today, drawn end to end: the people, the two faces of the app, the server, the database and the outside services. Then the six problems that limit scale, how we compare with the ctmsgas.com benchmark, and a phased plan to get to a branded, scalable product.

Live: `cylvero.vercel.app` Code: `Savage-Shaurya/Oxygen-Cylinder-Management`

Editable map: `docs/system-design.excalidraw`

TODAY, BY THE NUMBERS

What is already built

6 roles	2 app faces: Simple and Office	9 picture jobs in Simple mode	10 Office pages, 40+ forms
37 server commands with rules	13 API routes	7 database tables	160 Sarvam voice clips
191 automated tests, 0 failing			

Edit the map with the team. Open `docs/system-design.excalidraw` from GitHub at `excalidraw.com` (menu → Open). Every box and arrow is movable.

Regenerate after big changes. Run `node scripts/system-map.mjs`. The images below come from that same file, exported by Excalidraw.

Images for slides. `docs/system-design.png` (high resolution) and

docs/system-design.svg sit next to the map file.

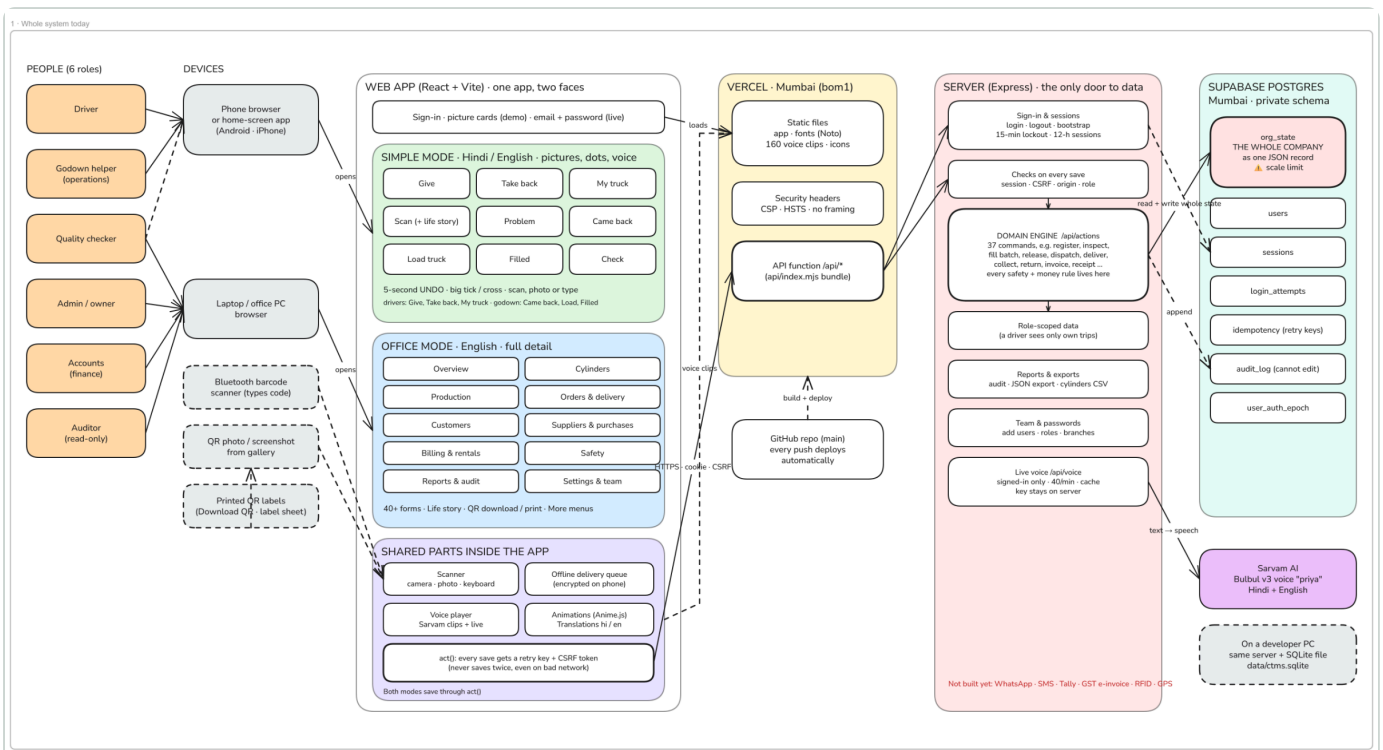
THE MAP, SECTION BY SECTION

How it works, end to end

1

Whole system today

People on phones and laptops use one web app. Phones open Simple mode, laptops open Office mode. Every save goes through one guarded door on the server, and all company data sits in one private Postgres schema in Mumbai.

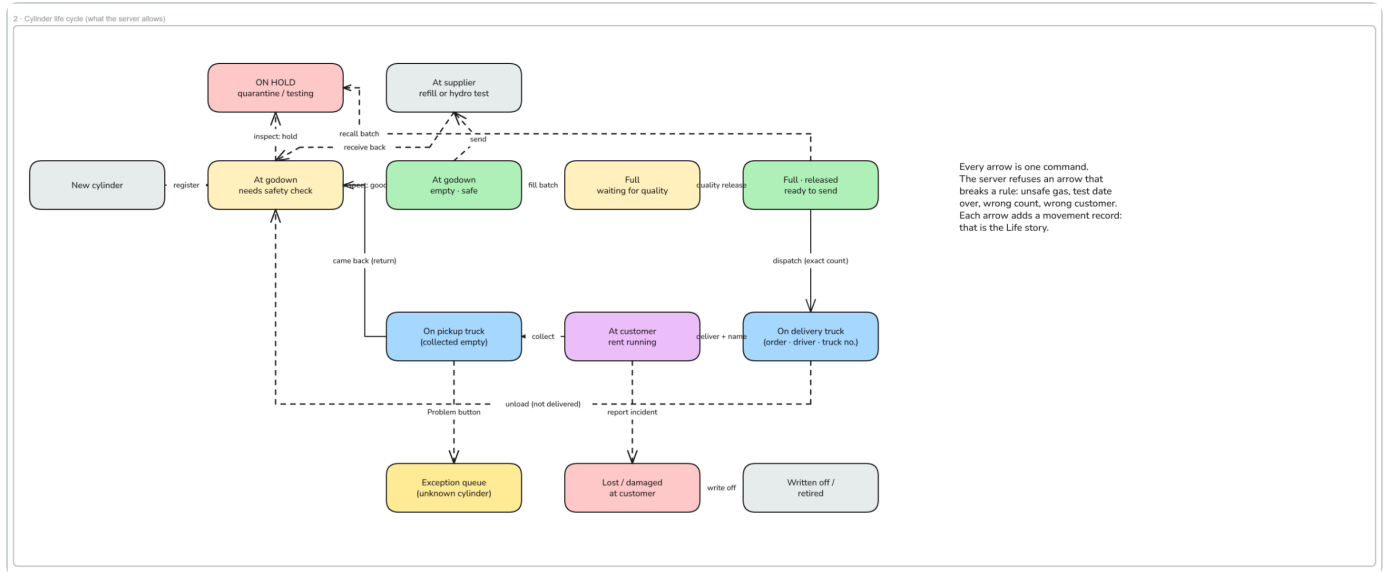


Red box on the right: the company's data is one JSON record. Finding 1 below.

2

Cylinder life cycle

Every arrow is one server command, and the server refuses any arrow that breaks a safety or money rule. Each arrow adds a movement record, which is what the Life story shows.

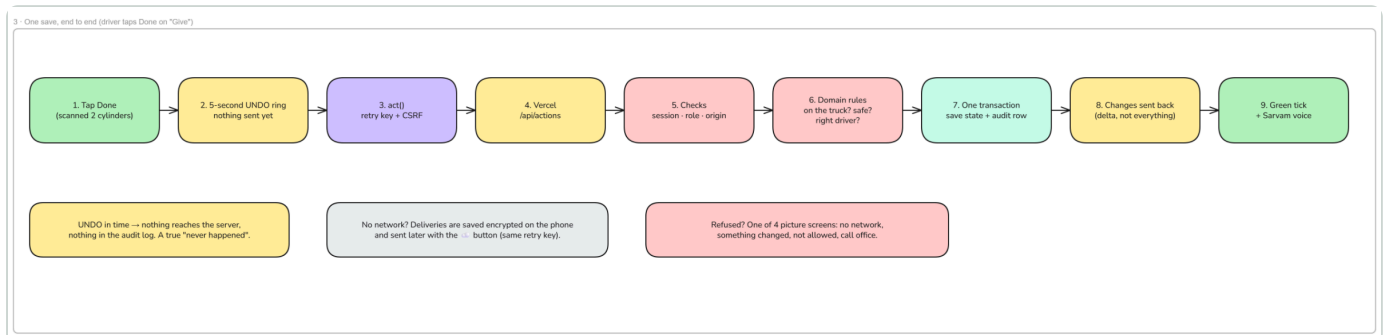


Solid arrows: the normal loop. Dashed arrows: exceptions.

3

One save, end to end

What happens between a driver tapping Done and hearing "2 cylinders given": the undo window, the retry key, the server checks, one transaction, and the reply.



Undo means nothing was sent, so nothing is in the audit log.

4

Who sees what

The server decides what each role receives and may do. Hiding a button is never the security.

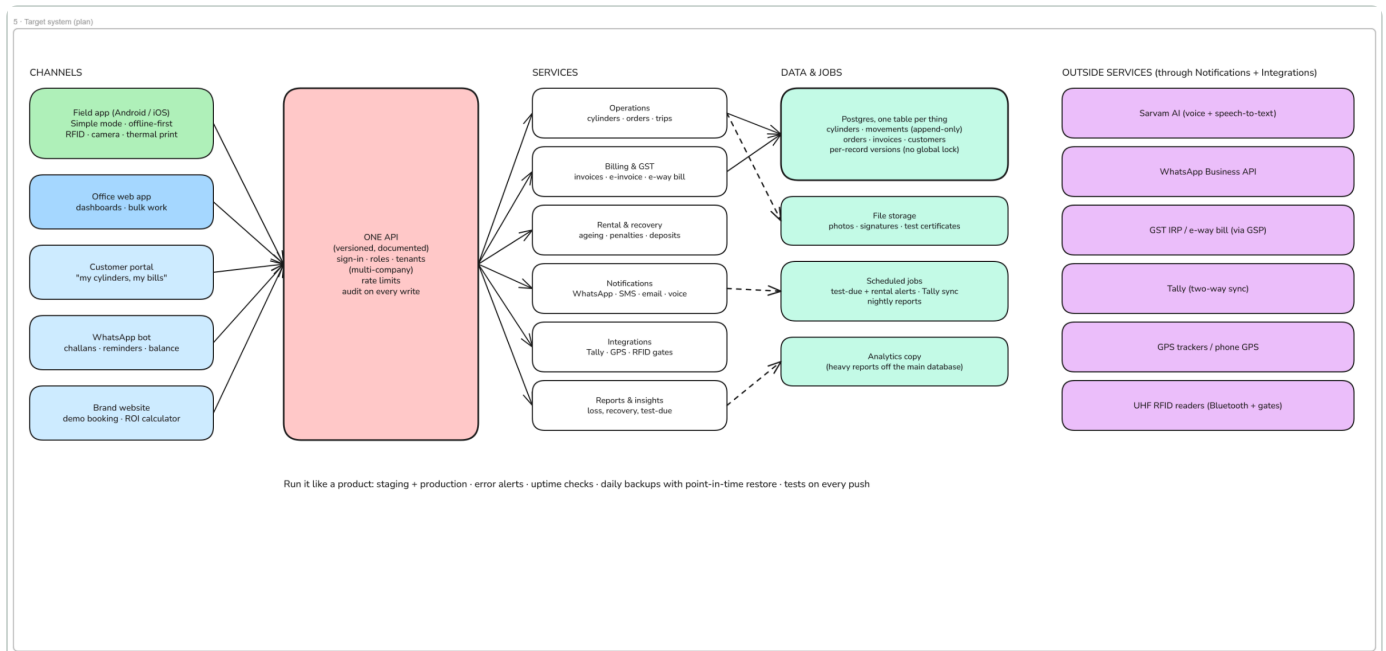


Drivers never reach Office mode.

5

Target system

Where we are heading: several channels on one documented API, services split by job, one table per kind of record, scheduled jobs for alerts, and the integrations Indian gas businesses expect.



Built in phases. See the roadmap below.

6

Against the benchmark

ctmsgas.com's published feature list, checked line by line against what we have.

6 - Against the benchmark (ctmsgas.com feature list)		
WE HAVE IT	PARTLY	MISSING
QR tagging + label print / download	Test-due tracking (no reminders yet)	WhatsApp challans / invoices / reminders
Barcode + Bluetooth scanner input	Hydro test (via supplier send / receive)	SMS + email alerts
Cylinder history (Life story)	Offline use (deliveries only)	Tally sync
Full lifecycle with safety rules	Dashboards (overview, not drill-down)	GST e-invoice (IRN) + e-way bill
Orders, dispatch, delivery challan	Customer ledger (inside Billing)	RFID readers + gates
Customer returns + pickups	Approval workflow (fill ≠ release)	GPS vehicle tracking, route planning
Fill batches + quality release / recall	Proof of delivery (typed name only)	Native Android / iOS app + thermal print
Rental billing, free days, deposits	Role management (6 fixed roles)	Photo + signature proof of delivery
GST invoices, receipts, credit notes		Customer portal, vendor portal
Multi-branch, 6 roles, audit log		Multi-company tenancy
CSV import, exports		Quotations, debit notes, penalties
Phone app (installable web app)		Liquid / line loss, yield reports

OURS ONLY: Hindi + English picture mode for workers who cannot read · Sarvam voice · 5-second undo · plain-language Life story · read QR from a photo · server-enforced safety rules

Details in the table below.

WHAT LIMITS SCALE

Six findings, most serious first

The app is safe and correct for a demo and a single small business. These are the things that stop it at lakhs of cylinders, many branches or many companies.

CRITICAL

The whole company is one JSON record

Every cylinder, order and movement lives inside one `org_state` row, read and rewritten on every save. Fine at 84 cylinders. At 5,00,000 cylinders and crores of movements, saves slow down and the row becomes too big to handle. Fix: one table per kind of record, with movements as an append-only table.

HIGH

One version number for the whole company

Two people saving different things at the same moment can get "something changed, check again". On a busy dispatch morning that becomes frequent. Fix: a version per record, so only real clashes conflict.

HIGH

Office browsers download everything at sign-in

Admin and office roles receive the full data set in one go. That is slow on 4G and will not fit in a phone's memory at scale. Fix: search, filters and pages answered by the server.

MEDIUM

Nothing runs on a schedule

Test-due dates and rental ageing are stored, but nobody is reminded. Fix: scheduled jobs for alerts, reminders, Tally sync and nightly reports.

MEDIUM

No place for photos or signatures

Proof of delivery is a typed name, and test certificates are a reference number. Fix: private file storage (Supabase Storage) linked to deliveries and cylinders.

MEDIUM

One company, demo sign-in

The data model already carries a company id, but there is one company and a public demo password. A real launch needs company sign-up, staff invites, and a simple PIN or badge sign-in for workers (after a security review).

BENCHMARK

ctmsgas.com and us, feature by feature

ctmsgas.com sells a cylinder ERP to Indian industrial, medical and LPG gas companies. Its pitch is QR and RFID tracking, Tally and WhatsApp integration, GST billing, rental ageing and Android and iOS apps. We are ahead on the field worker's experience and behind on integrations and hardware.

AREA	THEIR FEATURE	US	NOTE
Tracking	QR, barcode, history, lifecycle	Have	Plus photo-QR upload and plain-language Life story
Tracking	RFID readers and gates	Missing	Needs a native app and a reader vendor
Stock	Plant, vehicle, customer, branch stock	Have	Custody model covers all four

AREA	THEIR FEATURE	US	NOTE
Sales	Orders, delivery challans, invoices	Have	Quotations and debit notes missing
Finance	GST invoices, credit notes	Have	Integer-paise arithmetic
Finance	GST e-invoice (IRN), e-way bill	Missing	Through a GST Suvidha Provider
Finance	Tally two-way sync	Missing	Most asked-for by accountants
Rental	Rental billing, free days, deposits	Have	Penalties and ageing reminders missing
Quality	Test due, hydro test, maintenance	Partly	Rules enforced, no reminders yet
Production	Batches and production	Have	Fill, release, recall, reject
Production	Liquid loss, line loss, yield	Missing	Needs plant meter inputs
Messaging	WhatsApp challans, invoices, reminders; SMS; email	Missing	WhatsApp Business API provider needed
Field	Android and iOS apps, offline	Partly	Installable web app; offline for deliveries only
Field	Signature, photo proof, thermal printing	Missing	Comes with the native app
Logistics	GPS tracking, route planning	Missing	Driver assignment exists
Platform	Multi-branch, roles, audit logs	Have	Append-only audit, separation of duties
Platform	Multi-company, customer and vendor portals	Missing	Planned in phase 2 and 3

Where we already lead

- Picture-first Simple mode for workers who cannot read
- Hindi and English with natural Sarvam voice

- 5-second undo before anything is recorded
- Read a QR from a gallery photo
- Plain-language Life story for every cylinder
- Safety rules enforced on the server, not in the screen

Brand caution: "CTMS" is their product name, and they publish trademark guidelines. Our code folder is called CTMS, but the product should launch only as Cylvero (or another name we own), never reuse their wording, and never show customer numbers we have not earned.

ROADMAP

The plan, in five phases

Weeks are rough estimates for a small team of two to three developers. Phases 3 and 4 can overlap once phase 2 is done.

1. Brand and foundations 2–3 weeks

- Settle the name, run a trademark search, buy the domain, design the logo and colours.
- Turn the app's styles into one design system shared by the app and the website.
- Brand website with a demo-booking form and an ROI calculator.
- Separate staging and production; error alerts, uptime checks and point-in-time backups.

Done when the brand is consistent everywhere and a bad release can be caught on staging and rolled back.

2. Scale the core 4–6 weeks

- Move from the single JSON record to proper tables, with a one-time migration and the same rules and tests.
- Movements become an append-only table that the database itself protects, like the audit log.
- A version per record, server-side search and pages, and multi-company sign-up with staff invites.
- File storage for photos, signatures and certificates; a scheduled-job runner.

Done when a load test with 5,00,000 cylinders and 50 concurrent users keeps every save under one second.

3. Must-have integrations 6–8 weeks

- WhatsApp challans, invoices and reminders; SMS and email alerts.
- Test-due and rental-ageing alerts, penalties, quotations and debit notes.
- GST e-invoice and e-way bill; Tally sync.
- Owner dashboards with drill-down; customer portal ("my cylinders, my bills").

Done when an accountant can close the month without re-typing anything into Tally.

4. Field app and hardware 6–8 weeks

- Android and iOS app wrapping Simple mode, offline for every job, not just deliveries.
- Signature and photo proof of delivery; Bluetooth thermal printing of challans.
- Bluetooth UHF RFID readers and fixed gates at the filling line; GPS trips.

Done when a driver works a full day with no network and everything syncs correctly in the evening.



5. Stand-out features ongoing

- Voice commands in Hindi with Sarvam speech-to-text ("दो सिलेंडर दिए").
 - Loss and recovery predictions: which customers are holding cylinders too long.
 - WhatsApp bot for customers: "how many cylinders do I have, what do I owe".
 - More Indian languages, starting with the first customers' regions.
- | Done when field workers prefer our app to paper, and owners can see losses fall month by month.

NEEDS THE OWNER

Decisions before phase 1

Name and brand

Keep Cylvero, or choose another name we can trademark and get a domain for.

First market

Medical oxygen, industrial gas or LPG. LPG brings different regulations and much higher volumes.

Native app timing

Build the Android app in phase 4 as planned, or earlier if RFID and printing are must-haves for the first customers.

Partners

Which WhatsApp provider, which GST provider, and which RFID reader vendor to pilot with.

Pricing

Per cylinder, per branch, or per user. This shapes the multi-company design in phase 2.

Pilot customer

One real business to run the demo with for a month before a wider launch.